Gokul Vishnu

Bangalore, India (Open to Relocate to Germany)

+91 74117 39517 gokulvishnu.contact@gmail.com Gokul Vishnu

Summary

RTL Design Engineer with 4.8 years of experience in FPGA, SoC, and ASIC projects across industrial and aerospace domains. Specializes in Verilog and VHDL based RTL, protocol implementation, and timing focused digital design for high integrity systems. Hands-on experience with AXI-based architectures, DDR data paths, and image sensor interfaces from concept to hardware validation. Experienced in simulation, synthesis, and lab bring up using ModelSim, Vivado, Libero, and Lattice Diamond toolchains. Collaborates closely with hardware and software teams to integrate designs, resolve issues early, and meet project milestones. Proficient in creating comprehensive documentation, such as Conceptual Design Documents (CDD), Module Requirements Documents (MRD), and Interface Control Documents (ICD), ensuring clarity and precision in design specifications.

Core Skills

- **Digital Design:** RTL design, FPGA RTL, ASIC RTL, SoC blocks
- **HDLs & Programming:** Verilog HDL, VHDL
- **Protocols & Buses:** AXI Stream, AXI Lite, AXI Full, MIL 1553B, SPI, UART, PCI, Ethernet
- **FPGA Tools:** Xilinx Vivado, ModelSim, Libero, Lattice Diamond
- **Devices & Interfaces:** Xilinx xc7z030fbg676-1, Lattice LFE2 20E, Microchip M2GL090T(S)-FG676
- **Methods & Docs:** CDD, MRD, ICD

Experience

Klok Systems India Pvt. Ltd. — **August 2023 – Present**

Senior Engineer — *Bangalore, India*

- Own RTL design for FPGA and ASIC modules, aligning protocol requirements with timing, area and interface goals.
- Develop RTL for Device DMA, MIL-1553B and Device Interface blocks, integrating cleanly with system-level buses.
- Support simulation of PCI Interface RTL for functional checks and refined code using coverage report analysis.
- Run synthesis to review warnings, updated RTL to close issues and maintained CDD, MRD and ICD design documents.

Capgemini Engineering — **January 2023 – June 2023**

Software Engineer Lead — *Bangalore, India*

- Simulated RTL behavior against Module Requirements Documents to confirm logic and clock margin targets.
- Performed coverage report analysis for complex RTL blocks, closing gaps and improving verification quality metrics.
- Supported source code reviews against MRD, ensuring RTL implementation met functional and quality requirements.
- Worked with cross-functional design and verification teams to align RTL updates with system integration milestones.

Swan Sorter Systems Pvt. Ltd. — **December 2020 – January 2023**

Hardware Design Engineer — *Bangalore, India*

- Designed RTL for color sorting algorithm modules on Xilinx FPGAs to support high-throughput inspection.
- Developed DDR data acquisition with AXI DMA to move sensor streams while minimizing processor load.
- Programmed Toshiba image sensors and validated data paths from sensor to memory and sorting logic.
- Delivered FPGA designs and integrations for hardware testing while meeting schedule and quality targets.

Projects

GenericPCI Bridge – Collins Aerospace (Client) **February 2024 – Present**

- Owned CDD/MRD for Device DMA, Device Interface, and PCI Interface (memory-mapped and burst modes).
- Implemented RTL for Device Interface and Device DMA (memory-mapped and burst mode paths).
- Developed CDD, MRD and RTL for Device Bus arbitration, fixed PCI Interface Custom Source Code and MRD issues.

MIL 1553B Protocol – Klok Systems **August 2023 – January 2024**

- Authored the Interface Control Document (ICD) defining RT register map, MIL-1553B interface signals and timing.
- Prepared the Module Requirements Document capturing RT behaviour, protocol flows, test hooks and constraints.
- Designed RTL for Manchester encoder/decoder plus logic to split BC command and mode-code words into address fields.

MRD and Source Code Support – Capgemini Engineering **February 2023 – April 2023**

- Simulated RTL modules against MRD to confirm functional intent and clock domain constraints.
- Performed coverage report analysis for complex RTL designs, identifying gaps and driving additional test scenarios.
- Supported source code reviews against MRD, ensuring RTL implementation met functional and quality requirements.

Color Sorting Algorithm – Swan Sorter Systems Pvt. Ltd. **May 2022 – December 2022**

- Designed RTL to receive HMI data and apply flat-line correction on incoming sensor pixels for accurate processing.
- Implemented RTL to convert RGB inputs to HSL, enabling reliable color classification in the sorting workflow.

Data Acquisition to DDR Memory – Swan Sorter Systems Pvt. Ltd. **July 2021 – April 2022**

- Developed AXI DMA based block design to stream high-speed sensor data reliably into DDR memory buffers.
- Created custom RTL to synchronize Zynq processor and FPGA domains, ensuring coherent, low-latency data transfers.

Sensor Programming – Swan Sorter Systems Pvt. Ltd. **December 2020 – May 2021**

- Programmed Toshiba TCD2564DG and TCD1209DG image sensors to extract stable and usable image data.
- Coordinated with hardware teams to tune sensor settings for noise, exposure, and data integrity.

Key Achievements

- Completed complex RTL design ahead of schedule, improving manager satisfaction and driving repeat business.
- Achieved full pass rate in hardware validation for owned design modules against project requirements.
- Delivered key milestones under budget while improving design efficiency and overall functionality.

Education

East Point College of Engineering and Technology **August 2015 – June 2019**

Bachelor of Engineering in Electronics and Communication *Bangalore, India*

Certifications

- Professional Development Course in VLSI Design, Sandeepani School of VLSI Design, 2020.

Languages

- English – Full Professional Proficiency (C1).
- German – Intermediate Working Proficiency (B1).